Circular Motion (JEE LEVEL)

1. A disc is rotating in a room. A boy standing near the rim of the disc of radius 'R' finds the water droplet falling from the ceiling is always falling on his head. As one drop hits his head, other one starts from the ceiling. If height of the roof above his head is 'H', then ~~ang~~ find angular velocity of the disc

$$\left[\pi\sqrt{\frac{2g}{H}}\right]$$

2. A particle moves in a circle of radius ~~[illegible]~~ 2m with a speed given by $v = t^2$ m/s. Find net accl^n at $t = 2$ sec

$$\left[\sqrt{80}\ m/s^2\right]$$

3. A particle is projected with a speed 'u' at an angle 'θ' with the horizontal. Consider a small part of its path near the highest position. What is radius of curvature of this part of path (it approximates a circular arc)

$$\left[\frac{u^2\cos^2\theta}{g}\right]$$

4. A block of mass 'm' is kept on a horizontal ruler. The friction coefficient betn. the ruler and the block is 'μ'. The ruler is fixed at one end and the block is at a distance 'L' from the fixed end. The ruler rotates about the fixed end in horizontal plane. For what maximum angular speed the block does not slip.

$$\left[\sqrt{\frac{\mu g}{L}}\right]$$

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5. An insect of mass $m = 3kg$ is inside a vertical cylindrical drum of radius 2m that is rotating with an angular velocity of 5 rad/sec. The insect does not fall off. Find the minimum coeff of friction required.

[0.2]

6. A coin placed on a rotating turntable just slips if it is placed at a distance of 4cm from the centre. If the angular velocity of the turntable is doubled, then find the distance of it from centre at which it will just begin to slip? [1cm]

7. A car goes on a horizontal circular road of radius 'R'. Its speed increases at a constant rate $\frac{dv}{dt} = a$. The friction coefficient betn. the road and tyre is μ. Find the speed at which the car will skid.

$$\left[\left[(\mu^2 g^2 - a^2) R^2\right]^{1/4}\right]$$

8. A simple pendulum is suspended from the ceiling of a car taking a turn of radius 10m at a speed of 36 km/hr. Find angle made by the string of the pendulum with the vertical if this angle does not change during the turn. ($g = 10 m/s^2$)

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(9) A heavy particle hanging from a fixed point by a light inextensible string of length 'l' is projected horizontally with speed $\sqrt{5l}$. Find the inclination of the string to the vertical at the instant when tension in string is equal to weight of the particle. Also find speed of the particle at that instant.

$$\left[\cos^{-1}\left(\frac{2}{3}\right);\ \sqrt{\frac{gl}{3}}\right]$$

(10) A simple pendulum has length 'l' & mass of the bob is 'm'. The bob oscillates in a vertical circle. When the string makes angle 'α' to vertical speed of bob is 'v'. Find tension in string and magnitude of net force on the bob at that instant

$$\left[T = m\left(g\cos\theta + \frac{v^2}{l}\right)\right.$$

$$\left.F_{net} = m\sqrt{g^2\sin^2\alpha + \frac{v^4}{l^2}}\right]$$

(11) Find magnitude of average acc^n of the tip of the second hand of a clock in 10 sec. [length of second hand = 0.1 m]

$$\left(\frac{\pi}{3000}\ m/s^2\right)$$

(12) A particle is moving along a circular path of radius 5m. and with uniform speed 5m/s. What will be the average acc^n when the particle completes half revolution $\left(\frac{10}{\pi}\ m/s^2\right)$

(13) A pendulum of length $l = 1m$ is released from $\theta_0 = 60°$. Find rate of change of speed of the bob at $\theta = 30°$ $(g = 10\ m/s^2)$

[Ans $5\ m/s^2$]

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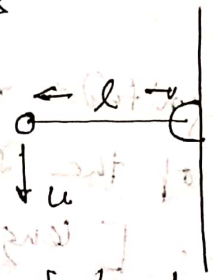
14. A particle is revolving with a constant angular acc^n 'α' in a circular path of radius 'r'. Find the time when the centripetal acceleration will be numerically equal to the tangential acceleration.

$\left[t = \frac{1}{\sqrt{\alpha}}\right]$

15. A bullet is moving horizontally with certain velocity. It pierces two paper discs rotating coaxially with angular speed 'ω' separated by a distance 'l'. If the hole made by the bullet on 2nd disc is shifted through an angle 'θ' with respect to that in the first disc, then find the of the bullet (assume constant)

$\left[\omega l/\theta\right]$

16. A ball of mass 'm' is attached to one end of a light rod of length l, the other end of it is hinged.



What minimum velocity 'u' should be imparted to the ball downward, so that it can complete the circle

$\left[\sqrt{39l}\right]$

17. A particle travels two and a half revolutions of the circle of radius R in time 't'. Find ratio of the average speed to the magnitude of average velocity of the particle in this time interval.

$\left[5\pi/2\right]$

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18. A particle slides on the surface of a fixed smooth sphere starting from the topmost point. Find the angle rotated by the radius through the particle, when it leaves contact with the sphere

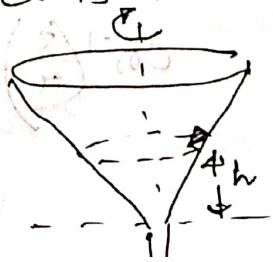
$\left[\cos^{-1}\left(\frac{2}{3}\right)\right]$

19. The bob of a stationary pendulum is given a sharp hit to impart it a horizontal speed $\sqrt{3gl}$. Find the angle rotated by the string before it becomes slack

$\left[\cos^{-1}\left(-\frac{1}{3}\right)\right]$

20. A hemispherical bowl of radius R is set rotating about its axis of symmetry which is kept vertical. A small block kept in the bowl rotates with the bowl without slipping on its surface. If the surface of the bowl is smooth and the angle made by the radius through the block with the vertical is θ. Find the angular speed at which the bowl is rotating. $\left[\omega = \sqrt{\frac{g \sec\theta}{R}}\right]$

21. A particle describes a horizontal circle on the smooth inner surface of a conical funnel (fig). If the plane of circle above vertex is 'h' Find speed of the particle? $(\sqrt{gh})$



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12 A particle is rotated in a vertical circle by connecting it to a string of length 'l' and keeping the other end of the string fixed. The minimum speed of the particle when the string is horizontal for which the particle will complete the circle is

a) $\sqrt{9l}$ b) $\sqrt{25l}$ c) $\sqrt{35l}$ d) $\sqrt{53l}$.

23. A stone tied to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instt of time the stone is at lowest position & has a speed 'u'. Find the magnitude of the change in velocity as it reaches a position, where the string is horizontal. $\sqrt{2(u^2-gL)}$

24. A particle of mass 'm' is moving in a circular path of constant radius 'r' such that its centripetal accelⁿ a_c is varying with time as $a_c = k^2 r t^2$. (k = const). Find power of the forces actg on the particle ($P = m k^2 r^2 t$)

25. A ball rests on a smooth solid sphere. If velocity given at topmost point is $\frac{\sqrt{gR}}{2}$ then find the angle θ where the ball will lose contact with the sphere? $\cos^{-1}\left(\frac{3}{4}\right)$

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